IN ACTION Federal Aviation Administration Uses Simulation to Solve Assignment Problem

The Federal Aviation Administration (FAA) is responsible for managing transportation in the air. This typically involves the assignment of airline flights to particular air-traffic routes in real time. On the surface, this problem may seem rather mundane. However, as air traffic demand has grown in recent years, the number of available air-traffic routes has diminished at any given time. This can make the associated assignment problem very hard. Confounding the problem is the weather, as thunderstorms can wreak havoc on the availability of air-traffic routes at any given time.

In 2005, the FAA developed a simulation decision-making tool known as the Airspace Flow Program (AFP) at a total cost of about $5 million. The AFP integrates upcoming and current flight data with impending storm weather data and simulates several different possible assignment decisions. All of these "look-ahead" simulations are then analyzed, allowing FAA decision makers to "pre-pick" a robust set of assignment solutions that minimize systemwide flight delays. The result is faster, more efficient flights for the traveler and hundreds of millions of dollars of annual savings for the airlines.

Source: Based on V. Sud, M. Tanino, J. Wetherly, M. Brennan, M. Lehky, K. Howard, and R. Oiesen, "Reducing Flight Delays Through Better Traffic Management," *Interfaces* 39, 1 (2009): 35–45, © Trevor S. Hale.

13.4 Simulation of a Queuing Problem

An important area of simulation application has been in the analysis of waiting line problems. As mentioned earlier, the assumptions required for solving queuing problems analytically are quite restrictive. For most realistic queuing systems, simulation may actually be the only approach available. This section illustrates the simulation at a large unloading dock and its associated queue.

Port of New Orleans

Barge arrivals and unloading rates are both probabilistic variables. Unless they follow the queuing probability distributions of Chapter 12, we must turn to a simulation approach.

Fully loaded barges arrive at night in New Orleans following their long trips down the Mississippi River from industrial midwestern cities. The number of barges docking on any given night ranges from 0 to 5. The probability of 0, 1, 2, 3, 4, or 5 arrivals is displayed in Table 13.9. In the same table, we establish cumulative probabilities and corresponding random number intervals for each possible value.

A study by the dock superintendent reveals that because of the nature of their cargo, the number of barges unloaded also tends to vary from day to day. The superintendent provides information from which we can create a probability distribution for the variable *daily unloading rate* (see Table 13.10). As we just did for the arrival variable, we can set up an interval of random numbers for the unloading rates.

Barges are unloaded on a first-in, first-out basis. Any barges that are not unloaded the day of arrival must wait until the following day. Tying up a barge in dock is an expensive

TABLE 13.9 Overnight Barge Arrival Rates and Random Number Intervals

NUMBER OF ARRIVALS	PROBABILITY	CUMULATIVE PROBABILITY	RANDOM NUMBER INTERVAL
0	0.13	0.13	01 to 13
1	0.17	0.30	14 to 30
2	0.15	0.45	31 to 45
3	0.25	0.70	46 to 70
4	0.20	0.90	71 to 90
5	0.10	1.00	91 to 00
	1.00		

TABLE 13.10 Unloading Rates and Random Number Intervals

DAILY UNLOADING RATE	PROBABILITY	CUMULATIVE PROBABILITY	RANDOM NUMBER INTERVAL
1	0.05	0.05	01 to 05
2	0.15	0.20	06 to 20
3	0.50	0.70	21 to 70
4	0.20	0.90	71 to 90
5	0.10	1.00	91 to 00
	1.00		

proposition, and the superintendent cannot ignore the angry phone calls from barge line owners reminding him that "Time is money!" He decides that before going to the Port of New Orleans's controller to request additional unloading crews, a simulation study of arrivals, unloadings, and delays should be conducted. A 100-day simulation would be ideal, but for purposes of illustration, the superintendent begins with a shorter 15-day analysis. Random numbers are drawn from the top row of Table 13.4 to generate daily arrival rates. They are drawn from the second row of Table 13.4 to create daily unloading rates. Table 13.11 shows the day-by-day port simulation.

TABLE 13.11 Queuing Simulation of Port of New Orleans Barge Unloadings

(1) DAY	(2) NUMBER DELAYED FROM PREVIOUS DAY	(3) RANDOM NUMBER	(4) NUMBER NIGHTLY ARRIVALS	(5) TOTAL TO BE UNLOADED	(6) RANDOM NUMBER	(7) NUMBER UNLOADED
1	(–)[a]	52	3	3	37	3
2	0	06	0	0	63	(0)[b]
3	0	50	3	3	28	3
4	0	88	4	4	02	1
5	3	53	3	6	74	4
6	2	30	1	3	35	3
7	0	10	0	0	24	(0)[c]
8	0	47	3	3	03	1
9	2	99	5	7	29	3
10	4	37	2	6	60	3
11	3	66	3	6	74	4
12	2	91	5	7	85	4
13	3	35	2	5	90	4
14	1	32	2	3	73	(3)[d]
15	0	00	5	5	59	3
	20		41			39
	Total delays		Total arrivals			Total unloadings

[a] We can begin with no delays from the previous day. In a long simulation, even if we started with 5 overnight delays, that initial condition would be averaged out.

[b] Three barges *could* have been unloaded on day 2. But because there were no arrivals and no backlog existed, zero unloadings took place.

[c] The same situation as noted in footnote b takes place.

[d] This time 4 barges could have been unloaded, but since only 3 were in the queue, the number unloaded is recorded as 3.

The superintendent will probably be interested in at least three useful and important pieces of information:

Here are the simulation results regarding average barge delays, average nightly arrivals, and average unloadings.

$$\text{Average number of barges delayed to the next day} = \frac{20 \text{ delays}}{15 \text{ days}}$$

$$= 1.33 \text{ barges delayed per day}$$

$$\text{Average number of nightly arrivals} = \frac{41 \text{ arrivals}}{15 \text{ days}} = 2.73 \text{ arrivals}$$

$$\text{Average number of barges unloaded each day} = \frac{39 \text{ unloadings}}{15 \text{ days}} = 2.60 \text{ unloadings}$$

When these data are analyzed in the context of delay costs, idle labor costs, and the cost of hiring extra unloading crews, it will be possible for the dock superintendent and port controller to make a better staffing decision. They may even elect to resimulate the process assuming different unloading rates that would correspond to increased crew sizes. Although simulation is a tool that cannot guarantee an optimal solution to problems such as this, it can be helpful in recreating a process and identifying good decision alternatives.

Using Excel to Simulate the Port of New Orleans Queuing Problem

Excel has been used to simulate the Port of New Orleans example, and the results are shown in Program 13.5. The VLOOKUP function is used as it was used in previous Excel simulations. Ten days of operation were simulated, and the results are displayed in rows 4 to 13 of the spreadsheet.

PROGRAM 13.5

Excel 2016 Model for Port of New Orleans Queuing Simulation

	A	B	C	D	E	F	G	H	I	J	K
1	**Port of New Orleans Barge Unloadings**										
2											
3	Day	Previously delayed	Random number	Arrivals	Total to be unoaded	Random Number	Possibly unloaded	Unloaded			
4	1	0	0.236949	1	1	0.434727	3	1			
5	2	0	0.229413	1	1	0.620976	3	1			
6	3	0	0.87566	4	4	0.212297	3	3			
7	4	1	0.827892	4	5	0.208373	3	3			
8	5	2	0.25807	1	3	0.282891	3	3			
9	6	0	0.407566	2	2	0.355181	3	2			
10	7	0	0.126246	0	0	0.040975	1	0			
11	8	0	0.184673	1	1	0.501543	3	1			
12	9	0	0.882713	4	4	0.596106	3	3			
13	10	1	0.091722	0	1	0.280506	3	1			
14											
15	**Barge Arrivals**						**Unloading rates**				
16	Demand	Probability	Lower	Cumulative	Demand		Number	Probability	Lower	Cumulative	Unloading
17	0	0.13	0	0.13	0		1	0.05	0	0.05	1
18	1	0.17	0.13	0.3	1		2	0.15	0.05	0.2	2
19	2	0.15	0.3	0.45	2		3	0.5	0.2	0.7	3
20	3	0.25	0.45	0.7	3		4	0.2	0.7	0.9	4
21	4	0.2	0.7	0.9	4		5	0.1	0.9	1	5
22	5	0.1	0.9	1	5						

KEY FORMULAS

	C	D
18	=C17+B17	=D17+B18

Copy C18:D18 to C19:D22

	I	J
18	=I17+H17	=J17+H18

Copy I18:J18 to I19:J21

	B	C	D	E	F	G	H
5	=E4-H4	=RAND()	=VLOOKUP(C5,$C$17:$E$22,3,TRUE)	=B5+D5	=RAND()	=VLOOKUP(F5,$I$17:K22,3,TRUE)	=MIN(E5,G5)

Copy B5:H5 to B6:H13

13.5 Simulation Model for a Maintenance Policy

Maintenance problems are an area in which simulation is widely used.

Simulation is a valuable technique for analyzing various maintenance policies before actually implementing them. A firm can decide whether to add more maintenance staff based on machine downtime costs and costs of additional labor. It can simulate replacing parts that have not yet failed in exploring ways to prevent future breakdowns. Many companies use computerized simulation models to decide if and when to shut down an entire plant for maintenance activities. This section provides an example of the value of simulation in setting maintenance policy.

Three Hills Power Company

The Three Hills Power Company provides electricity to a large metropolitan area through a series of almost 200 hydroelectric generators. Management recognizes that even a well-maintained generator will have periodic failures or breakdowns. Energy demands over the past 3 years have been consistently high, and the company is concerned over downtime of generators. It currently employs four highly skilled repairpersons at $30 per hour. Each works every fourth 8-hour shift. In this way, there is a repairperson on duty 24 hours a day, 7 days a week.

As expensive as the maintenance staff salaries are, breakdown expenses are even higher. For each hour that one of its generators is down, Three Hills loses approximately $75. This amount is the charge for reserve power that Three Hills must "borrow" from the neighboring utility company.

Stephanie Robbins has been assigned to conduct a management analysis of the breakdown problem. She determines that simulation is a workable tool because of the probabilistic nature of this problem. She decides her objective is to determine (1) the service maintenance cost, (2) the simulated machine breakdown cost, and (3) the total of these breakdown and maintenance costs (which gives the total cost of this system). Since the total downtime of the machines is needed to compute the breakdown cost, she must know when each machine breaks and when each machine returns to service. Therefore, a next-event-step simulation model must be used. In planning for this simulation, a flowchart, as seen in Figure 13.4, is developed.

Robbins identifies two important maintenance system components. First, the time between successive generator breakdowns varies historically from as little as 0.5 hour to as much as 3 hours. For the past 100 breakdowns, she tabulates the frequency of various times between machine failures (see Table 13.12). She also creates a probability distribution and assigns random number intervals to each expected time range.

Robbins then notes that the people who do repairs log their maintenance time in 1-hour time blocks. Because of the time it takes to reach a broken generator, repair times are generally rounded to 1, 2, or 3 hours. In Table 13.13, she performs a statistical analysis of past repair times, similar to that conducted for breakdown times.

Robbins begins conducting the simulation by selecting a series of random numbers to generate simulated times between generator breakdowns and a second series to simulate repair times required. A simulation of 15 machine failures is presented in Table 13.14. We now examine the elements in the table, one column at a time.

COLUMN 1: BREAKDOWN NUMBER. This is the count of breakdowns as they occur, going from 1 to 15.

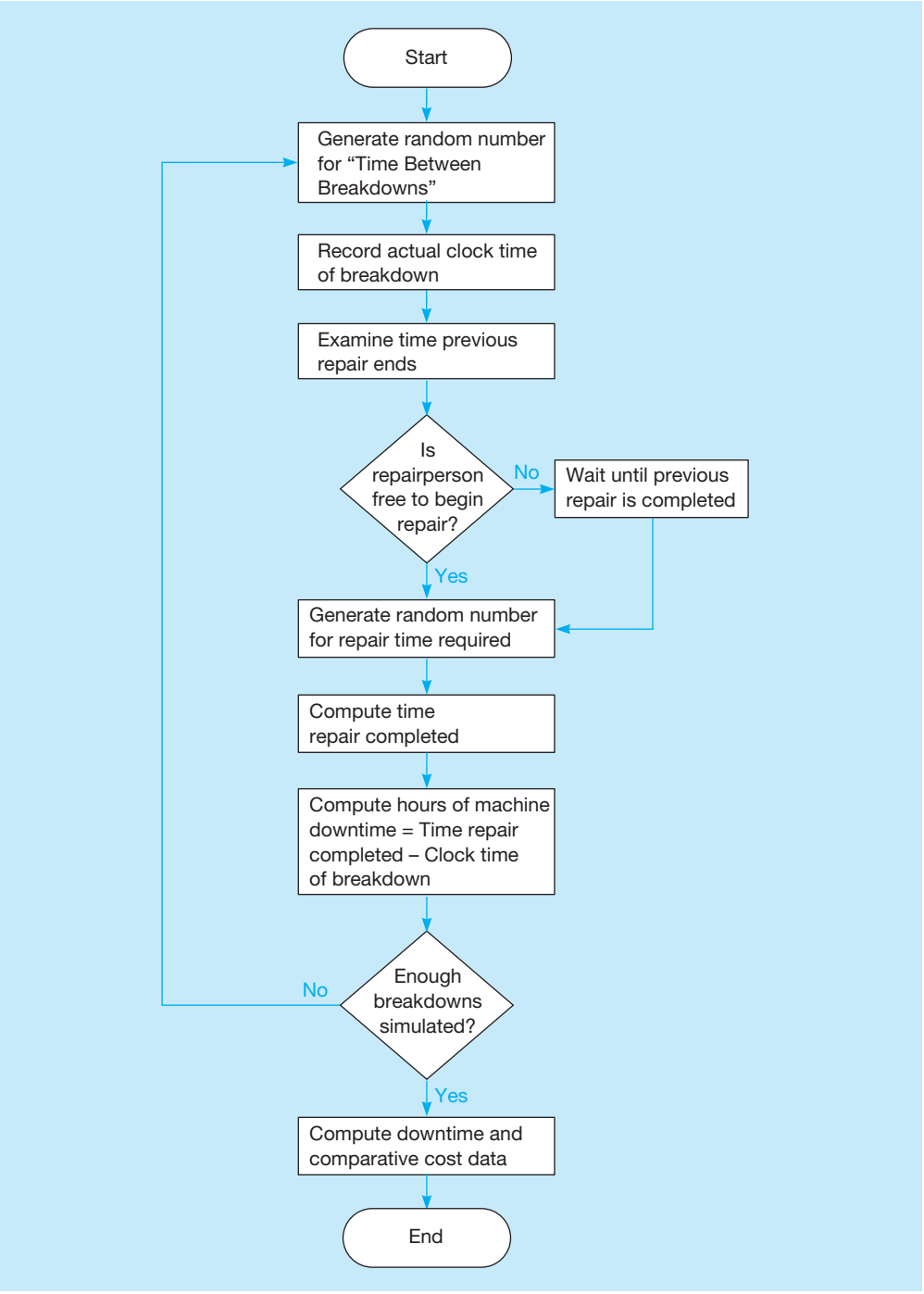
COLUMN 2: RANDOM NUMBER FOR BREAKDOWNS. This is a number used to simulate time between breakdowns. The numbers in this column have been selected from Table 13.4, from the second column from the right-hand side of the table.

COLUMN 3: TIME BETWEEN BREAKDOWNS. This number is generated from column 2 random numbers and the random number intervals defined in Table 13.12. The first random number, 57, falls in the interval 28 to 60, implying a time of 2 hours since the prior breakdown.

COLUMN 4: TIME OF BREAKDOWN. This converts the data in column 3 into an actual time of day for each breakdown. This simulation assumes that the first day begins at midnight (00:00 hours). Since the time between zero breakdowns and the first breakdown is 2 hours, the first recorded machine failure is at 02:00 on the clock. The second breakdown, you note, occurs 1.5 hours later, at a calculated clock time of 03:30 (or 3:30 A.M.).

COLUMN 5: TIME REPAIRPERSON IS FREE TO BEGIN REPAIR. This is 02:00 hours for the first breakdown if we assume that the repairperson began work at 00:00 hours and was not tied up from a previous generator failure. Before recording this time on the second and all subsequent lines, however,

FIGURE 13.4
Three Hills Flow Diagram



we must check column 8 to see what time the repairperson finishes the previous job. Look, for example, at the seventh breakdown. The breakdown occurs at 15:00 hours (or 3:00 P.M.). But the repairperson does not complete the previous job, the sixth breakdown, until 16:00 hours. Hence, the entry in column 5 is 16:00 hours.

One further assumption is made to handle the fact that each repairperson works only an 8-hour shift: when each person is replaced by the next shift, he or she simply hands the tools over to the new worker. The new repairperson continues working on the same broken generator until the job is completed. There is no lost time and no overlap of workers. Hence, labor costs for each 24-hour day are exactly 24 hours $\times$ \$30 per hour $=$ \$720.

COLUMN 6: RANDOM NUMBER FOR REPAIR TIME. This is a number selected from the rightmost column of Table 13.4. It helps simulate repair times.

TABLE 13.12
Time Between Generator Breakdowns at Three Hills Power

TIME BETWEEN RECORDED MACHINE FAILURES (HOURS)	NUMBER OF TIMES OBSERVED	PROBABILITY	CUMULATIVE PROBABILITY	RANDOM NUMBER INTERVAL
0.5	5	0.05	0.05	01 to 05
1.0	6	0.06	0.11	06 to 11
1.5	16	0.16	0.27	12 to 27
2.0	33	0.33	0.60	28 to 60
2.5	21	0.21	0.81	61 to 81
3.0	19	0.19	1.00	82 to 00
Total	100	1.00		

TABLE 13.13
Generator Repair Times Required

REPAIR TIME REQUIRED (HOURS)	NUMBER OF TIMES OBSERVED	PROBABILITY	CUMULATIVE PROBABILITY	RANDOM NUMBER INTERVAL
1	28	0.28	0.28	01 to 28
2	52	0.52	0.80	29 to 80
3	20	0.20	1.00	81 to 00
Total	100	1.00		

COLUMN 7: REPAIR TIME REQUIRED. This is generated from column 6's random numbers and Table 13.13's repair time distribution. The first random number, 07, represents a repair time of 1 hour, since it falls in the random number interval 01 to 28.

COLUMN 8: TIME REPAIR ENDS. This is the sum of the entry in column 5 (time repairperson is free to begin) plus the required repair time from column 7. Since the first repair begins at 02:00 and takes 1 hour to complete, the time repair ends is recorded in column 8 as 03:00.

COLUMN 9: NUMBER OF HOURS THE MACHINE IS DOWN. This is the difference between column 4 (time of breakdown) and column 8 (time repair ends). In the case of the first breakdown, that difference is 1 hour (03:00 minus 02:00). In the case of the tenth breakdown, the difference is 23:00 hours minus 19:30 hours, or 3.5 hours.

Cost Analysis of the Simulation

The simulation of 15 generator breakdowns in Table 13.14 spans a time of 34 hours of operation. The clock began at 00:00 hours of day 1 and ran until the final repair at 10:00 hours of day 2.

The critical factor that interests Robbins is the total number of hours that generators are out of service (from column 9). This is computed to be 44 hours. She also notes that toward the end of the simulation period, a backlog is beginning to appear. The thirteenth breakdown occurred at 01:00 hours but could not be worked on until 04:00 hours. The fourteenth and fifteenth breakdowns experienced similar delays. Robbins is determined to write a computer program to carry out a few hundred more simulated breakdowns but first wants to analyze the data she has collected thus far.

She measures her objectives as follows:

$$\begin{aligned}
\text{Service maintenance cost} &= 34 \text{ hours of worker service time} \times \$30 \text{ per hour} \\
&= \$1{,}020 \\
\text{Simulated machine breakdown cost} &= 44 \text{ total hours of breakdown} \\
&\quad \times \$75 \text{ lost per hour of downtime} \\
&= \$3{,}300 \\
\text{Total simulated maintenance cost of the current system} &= \text{Service cost} + \text{Breakdown cost} \\
&= \$1{,}020 + \$3{,}300 \\
&= \$4{,}320
\end{aligned}$$

TABLE 13.14 Simulation of Generator Breakdowns and Repairs

(1) BREAKDOWN NUMBER	(2) RANDOM NUMBER FOR BREAKDOWNS	(3) TIME BETWEEN BREAKDOWNS (HOURS)	(4) TIME OF BREAKDOWN	(5) TIME REPAIRPERSON IS FREE TO BEGIN THIS REPAIR	(6) RANDOM NUMBER FOR REPAIR TIME	(7) REPAIR TIME REQUIRED (HOURS)	(8) TIME REPAIR ENDS	(9) NUMBER OF HOURS MACHINE DOWN
1	57	2	02:00	02:00	07	1	03:00	1
2	17	1.5	03:30	03:30	60	2	05:30	2
3	36	2	05:30	05:30	77	2	07:30	2
4	72	2.5	08:00	08:00	49	2	10:00	2
5	85	3	11:00	11:00	76	2	13:00	2
6	31	2	13:00	13:00	95	3	16:00	3
7	44	2	15:00	16:00	51	2	18:00	3
8	30	2	17:00	18:00	16	1	19:00	2
9	26	1.5	18:30	19:00	14	1	20:00	1.5
10	09	1	19:30	20:00	85	3	23:00	3.5
11	49	2	21:30	23:00	59	2	01:00	3.5
12	13	1.5	23:00	01:00	85	3	04:00	5
13	33	2	01:00	04:00	40	2	06:00	5
14	89	3	04:00	06:00	42	2	08:00	4
15	13	1.5	05:30	08:00	52	2	10:00	4.5
								Total 44

IN ACTION Simulating Jackson Memorial Hospital's Operating Rooms

Miami's Jackson Memorial Hospital, Florida's largest hospital, with 1,576 inpatient beds, is also one of the finest hospitals in the United States. Department of Management Systems Engineering is constantly seeking ways of increasing hospital efficiency, and the construction of new operating rooms (ORs) prompted the development of a simulation of the existing 31 ORs.

The OR boundary includes the Holding Area and the Recovery Area, both of which were experiencing problems due to ineffective scheduling of OR services. A simulation study, modeled using the ARENA software package, sought to maximize the current use of OR rooms and staff. Inputs to the model included (1) the amount of time a patient waits in holding, (2) the specific process the patient undergoes, (3) the staff schedule, (4) room availability, and (5) time of day.

The first hurdle that the research team had to deal with at Jackson was the vast number of records to scour to extract data for the probabilistic simulation model. The second hurdle was the *quality* of the data. A thorough analysis of the records determined which were good and which had to be discarded. In the end, Jackson's carefully screened databases led to a good set of model inputs. The simulation model then successfully developed five measures of OR performance: (1) number of procedures a day, (2) average case time, (3) staff utilization, (4) room utilization, and (5) average waiting time in the holding area.

Source: Based on M. A. Centeno et al., "Challenges of Simulating Hospital Facilities," in *Proceedings of the 12th Annual Conference of the Production and Operations Management Society* (Orlando, FL: Production and Operations Management Society, March 2001), 50, © Trevor S. Hale.

A total cost of $4,320 is reasonable only when compared with other, more attractive or less attractive maintenance options. Should, for example, the Three Hills Power Company add a second full-time repairperson to each shift? Should it add just one more worker and let him or her come on duty every fourth shift to help catch up on any backlogs? These are two alternatives that Robbins may choose to consider through simulation. You can help by solving Problem 13.25 at the end of the chapter.

As mentioned at the outset of this section, simulation can also be used in other maintenance problems, including the analysis of *preventive maintenance*. Perhaps the Three Hills Power Company should consider strategies for replacing generator motors, valves, wiring, switches, and other miscellaneous parts that typically fail. It could (1) replace all parts of a certain type when one fails on any generator or (2) repair or replace all parts after a certain length of service based on an estimated average service life. This would again be done by setting probability distributions for failure rates, selecting random numbers, and simulating past failures and their associated costs.

Preventive maintenance policies can also be simulated.

BUILDING AN EXCEL SIMULATION MODEL FOR THREE HILLS POWER COMPANY Program 13.6 provides an Excel spreadsheet approach to simulating the Three Hills Power maintenance problem.

PROGRAM 13.6

Excel 2016 Model for Three Hills Power Company Maintenance Problem

	A	B	C	D	E	F	G	H	I	J	K
1	**Three Hills Power Company**										
2											
3	Breakdown number	Random number	Time between breakdowns	Time of breakdowns	Time repairperson is free	Random Number	Repair time	Repair ends			
4	1	0.0679	1	1	1	0.8716	3	4			
5	2	0.4740	2	3	4	0.9099	3	7			
6	3	0.4743	2	5	7	0.2604	1	8			
7	4	0.6149	2.5	7.5	8	0.8114	3	11			
8	5	0.1472	1.5	9	11	0.6365	2	13			
9	6	0.5319	2	11	13	0.5700	2	15			
10	7	0.5553	2	13	15	0.3763	2	17			
11	8	0.7095	2.5	15.5	17	0.3219	2	19			
12	9	0.9536	3	18.5	19	0.3988	2	21			
13	10	0.2717	2	20.5	21	0.9208	3	24			
14											
15	Demand Table						Repair times				
16	Time between breakdowns	Probability	Lower	Cumulative	Demand		Time	Probability	Lower	Cumulative	Lead time
17	0.5	0.05	0	0.05	0.5		1	0.28	0.00	0.28	1
18	1.0	0.06	0.05	0.11	1		2	0.52	0.28	0.80	2
19	1.5	0.16	0.11	0.27	1.5		3	0.20	0.80	1.00	3
20	2.0	0.33	0.27	0.6	2						
21	2.5	0.21	0.6	0.81	2.5						
22	3.0	0.19	0.81	1	3						

KEY FORMULAS

	C	D	E
18	=C17+B17	=D17+B18	=A18

Copy C18:E18 to C19:E22

	I	J	K
18	=I17+H17	=J17+H18	=G18

	B	C	D	E	F	G	H
5	=RAND()	=VLOOKUP(B5,$C$17:$E$22,3,TRUE)	=D4+C5	=MAX(D5,H4)	=RAND()	=VLOOKUP(F5,$I$17:$K$19,3,TRUE)	=E5+G5

Copy B5:H5 to B6:H13

13.6 Other Simulation Issues

Simulation is one of the most widely used tools in business. As we have seen in the earlier parts of this chapter, applications abound, as we are not restricted by the assumptions in some of the mathematical models discussed in earlier chapters. In this section, we look at a few other issues related to simulation, including some of the software tools available.

Two Other Types of Simulation Models

Simulation models are often broken into three categories. The first, the Monte Carlo method just discussed, uses the concepts of probability distribution and random numbers to evaluate system responses to various policies. The other two categories are operational gaming and systems simulation. Although in theory the three methods are distinctly different, the growth of computerized simulation has tended to create a common basis in procedures and blur these differences.[3]

OPERATIONAL GAMING **Operational gaming** refers to simulation involving two or more competing players. The best examples are military games and business games. Both allow participants to match their management and decision-making skills in hypothetical situations of conflict.

Military games are used worldwide to train a nation's top military officers, to test offensive and defensive strategies, and to examine the effectiveness of equipment and armies. Business games, first developed by the firm Booz Allen Hamilton in the 1950s, are popular with both executives and business students. They provide an opportunity to test business skills and decision-making ability in a competitive environment. The person or team that performs best in the simulated environment is rewarded by knowing that his or her company has been most successful in earning the largest profit, grabbing a high market share, or perhaps increasing the firm's trading value on the stock exchange.

During each period of competition, be it a week, month, or quarter, teams respond to market conditions by coding their latest management decisions with respect to inventory, production, financing, investment, marketing, and research. The competitive business environment is simulated by computer, and a new printout summarizing current market conditions is presented to the players. This allows teams to simulate years of operating conditions in a matter of days, weeks, or a semester.

SYSTEMS SIMULATION **Systems simulation** is similar to business gaming in that it allows users to test various managerial policies and decisions to evaluate their effect on the operating environment. This variation of simulation models the dynamics of large *systems*. Such systems include corporate operations,[4] the national economy, a hospital, or a city government system.

In a *corporate operating system*, sales, production levels, marketing policies, investments, union contracts, utility rates, financing, and other factors are all related in a series of mathematical equations that are examined by simulation. In a simulation of an *urban government*, systems

[3]Theoretically, random numbers are used only in Monte Carlo simulation. However, in some complex gaming or systems simulation problems in which all relationships cannot be defined exactly, it may be necessary to use the probability concepts of the Monte Carlo method.

[4]This is sometimes referred to as *industrial dynamics*, a term coined by Jay Forrester. Forrester's goal was to find a way "to show how policies, decisions, structure, and delays are interrelated to influence growth and stability" in industrial systems. See J. W. Forrester, *Industrial Dynamics* (Cambridge, MA: MIT Press, 1961).